

In 2023, the 7th of September, extreme rainfall turned the Wong Tai Sin subway into a churning river, a lone adult waded through waist-deep debris, desperately trying to carry her child to safety. That day, Hong Kong experienced its heaviest flood since 1884 (CNN, 2023). The flood injured hundreds, disabled inundated the ground floor of many commercial facilities and disabled the whole MTR system, as well as ground transportation. Paralysing the city for more than two days. Yet, this kind of unusual weather has become more and more common in recent years (IPCC 2023). Heavy flooding is not a rare incident anymore and if we don't do anything about it, it's only going to exacerbate at an even faster rate.

What's causing this drastic change in the rate of heavy flooding is a very simple mechanism: Global warming causes the atmosphere to expand and allow it to hold more water, this leads to less frequent, but much heavier rainfall (NASA, n.d.). What's even worse is that water vapour also acts as a greenhouse gas and traps heat within the atmosphere, creating a positive feedback loop that intensifies itself (NASA, n.d.). From the Hong Kong perspective, this could mean that on top of the yearly typhoons, we may also be experiencing yearly flooding in the near future, leading to the city being frequently paralysed. On a global scale. We could see cities being disrupted by heavy floods every year, the homes of many will be destroyed and the economy will be forced into a state of constant recovery .

To stop this issue from further exacerbating, we need to cut down greenhouse gas emissions, the cause of global warming. I propose that the government should impose a carbon tax on the use of fossil fuels, the primary contributor to greenhouse gas emissions (UN, n.d.)

However, simply increasing the cost of fossil fuels is not going to fully solve the problem. It could raise the cost of living and production, putting most of the burden on society. Because of this, the government should use the revenue from the carbon tax to subsidize green energy

infrastructure, specifically offshore wind turbines and solar panels, as these are more feasible options for Hong Kong. These new regulations would help transition electricity production from fossil-fuel-based energy to cleaner sources, which is a major step in the right direction.

On the social side, the government should also subsidize electric vehicles and expand green infrastructure. By installing rooftop gardens, roadside vegetation reduces the urban heat land effect (EPA, n.d.). With these regulations, society would slowly shift from fossil-fuel-based energy to cleaner electrical energy. As electricity becomes cleaner, overall carbon emissions would be reduced.

Making fossil fuel more expensive will make renewable energy a better alternative option. While this would not stop flooding immediately, it would slow down the rate of temperature increase. Since warmer air holds more water vapour and causes heavier rainfall, reducing emissions would lower the long-term risk of extreme rainfall and flooding (WMO, n.d.). If more regions implemented similar policies, greenhouse gas emissions could decrease significantly, and the trend of worsening floods could eventually be slowed or even reversed.

This transition will inevitably disrupt carbon-intensive industries in the short term, but as the transformation slowly takes place, more of the green energy infrastructures will be implemented, and with a more matured renewable energy generation system, the burden on the market will be eased and the market will be back to normal.

As a part of society, we should start by doing the small things in life to reduce the emission we produce. We can reduce our AC usage at home, unplugging devices when they are not being used. Choosing greener options for going out when time isn't a major concern, recycling waste, and reducing the consumption of red meat. These actions may seem small, but they still matter because they reduce energy demand and lower our personal carbon

footprint. If more people make these changes, the combined effect can help slow down the worsening of global warming.

Due to global warming, extreme flooding is becoming more severe and more frequent, and it will self intensify if no action is taken. But if we all take action in whatever way we can manage, and only if we all make an effort to it, meaningful change will be possible.

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